Free to Be Trusted? Organizational Constraints on Trust in Boundary Spanners

Vincenzo Perrone • Akbar Zaheer • Bill McEvily

The Institute of Organization and MIS, Bocconi University, and SDA, School of Management, via F. Bocconi n. 8, 20136 Milan, Italy

Strategic Management and Organization Department, Carlson School of Management, 3-365, University of Minnesota, 321 19th Avenue South, Minneapolis, Minnesota 55455

Graduate School of Industrial Administration, Carnegie Mellon University, Pittsburgh, Pennsylvania 15213-3890

vincenzo.perrone@uni-bocconi.it • azaheer@csom.umn.edu • bmcevily@andrew.cmu.edu

Abstract

We present a view of trust in boundary spanners as explained by the extent of role autonomy, a multidimensional concept that reflects the discretion that agents have in interpreting and enacting their roles. We argue that, in a buyer-supplier context, purchasing managers will be trusted to a greater extent by supplier representatives when they are free from constraints that limit their ability to interpret their boundary-spanning roles. We conceptualize and measure three key components of role autonomy: Functional influence, tenure, and clan culture. Taken together, these components of role autonomy shape and define the purchasing manager's willingness and capacity to make and uphold commitments to supplier representatives. Role autonomy permits purchasing managers to engage in discretionary behaviors that allow supplier representatives to learn about their underlying motives and intentions. We test hypotheses linking the components of role autonomy to trust on a sample of 119 buyer-supplier relationships. We use a dyadic research design that combines data from purchasing managers and supplier representatives. The results suggest that granting purchasing managers greater autonomy enhances supplier representative trust in purchasing managers. By drawing attention to role autonomy as a feature of organizations that influences trust we highlight the importance of organizational context in contributing to a deeper understanding of trust.

(*Trust*; *Role Autonomy*; *Organizational Context*; *Buyer-Supplier Relationships*; *Boundary Spanner*)

One of the most difficult decisions to make is whether to trust another person. While trust in individuals has been rightly understood as being affected by individual characteristics, interpersonal interactions, and institutional environments, in an organizational context the decision is further complicated (Blau 1964, Fukuyama 1995, Zucker 1986). Because the structures, processes, and culture of an organization all shape the behavior of its members and influence their interactions, making attributions about the motives and intentions behind individual behavior is not straightforward. Rather, assessing the trustworthiness of individual organizational members involves considering the effects of organizational context on individual behavior.

One way that organizational context affects individual behavior is through roles (Shapiro 1987). Since roles constrain behavior, there is reason to believe that roles may influence the degree of trust placed in agents performing roles. The precise effect of roles on trust remains unclear, however. From one perspective, organizationally defined role constraints may make the behavior of agents more consistent and consequently more trustworthy (Barber 1983). From another perspective, role constraints make it more difficult to ascertain the trustworthiness of organizational agents because role constraints limit the ability to make attributions about the motives and intentions of an agent *outside* of their role (Ring and Van de Ven 1994).

However, both of these perspectives are premised on the assumption that agent behavior is fully dictated by the role. In contrast, we argue for a more nuanced view of roles as emergent and malleable. A role is not simply a set of rules that are slavishly adhered to. Rather than being a straitjacket, we maintain that roles are often open to wide latitude in interpretation and are frequently subject to multiple and conflicting pressures (Kahn et al. 1964). Consequently, we cannot predict trust merely based on formal role definition because organizational agents are neither completely constrained nor completely free to be trusted. Accordingly, understanding the subtleties of how roles influence the level of trust placed in organizational agents is an important research question.

One role that is particularly interesting and important with regard to trust is that of the boundary spanner.

Organization Science © 2003 INFORMS
Vol. 14, No. 4, July–August 2003, pp. 422–439

1047-7039/03/1404/0422$05.00
1526-5455 electronic ISSN

Boundary spanners process the information coming from the partner organization, represent the interests of their own firm in the relationship, and "link organizational structure to environmental elements" (Aldrich and Herker 1977, p. 218; Adams 1976). Due to their unique location in the organization, boundary spanners are simultaneously exposed to competing expectations from their own and from the partner organization (Kahn et al. 1964, Strauss 1962, Spekman 1979).[1] In dealing with these conflicting influences, boundary spanners are frequently required to make choices and take actions beyond their role as they confront unplanned contingencies that cannot be specified in their role a priori. In the course of handling these contingencies, boundary spanners exercise individual discretion, which provides others with information about their trustworthiness. We are interested in understanding how the definition of the boundary-spanning role affects the degree of discretion available to the organizational agents performing those roles and the trust reposed in them.

More specifically, we focus on the level of trust placed in boundary-spanning agents within the context of buyer-supplier relationships. Our objective is to carefully examine how role autonomy—the leeway agents have in interpreting and enacting their roles—influences the level of trust placed in the boundary spanner. In particular, we examine supplier representatives' trust in purchasing managers.[2] We propose that the level of trust supplier representatives place in purchasing managers is substantially influenced by the degree of autonomy purchasing managers have in interpreting their boundary-spanning roles. Role autonomy influences trust in purchasing managers by creating opportunities to meet the expectations of supplier representatives. Drawing on role theory (Biddle 1979, Seligman 1997), we build a multidimensional conceptualization of role autonomy and argue that the discretionary behaviors enabled by role autonomy help shape the level of trust placed in purchasing managers by supplier representatives. By testing our theory with a sample of 119 buyer-supplier relationships, we contribute to the expanding, yet still sparse, empirical literature on trust. We further contribute by redirecting attention from interpersonal effects toward "impersonal" organizational effects on trust.

Theory and Hypotheses

Trust in the Purchasing Manager

Within the context of this study, we define trust as a supplier's expectation that a purchasing manager (1) can be relied on to fulfill obligations (Anderson and Weitz 1989), (2) will behave in a predictable manner, and (3) will act and negotiate fairly when the possibility for opportunism is present (Bromiley and Cummings 1995, Van den Bos et al. 1998).[3] We therefore conceptualize trust in terms of three core components: Reliability, predictability,[4] and fairness.[5] Viewing trust as an expectation highlights the inherent uncertainty surrounding a purchasing-manager's behavior. Moreover, this definition reflects the notion that trust includes a leap of faith (Lewis and Weigert 1985) because the supplier cannot know with absolute certainty whether the purchasing manager will fulfill both explicit and implicit expectations. In this way, our definition of trust is largely consistent with other mainstream conceptualizations that view trust as encompassing more than merely the predictability of another (Lewicki and Bunker 1996).

The notion of trust as an expectation is also consistent with the view of trust as a psychological state of the trustor (Rousseau et al. 1998), which is distinct from trustworthiness (Butler 1991, Williams 2001) as a latent characteristic of the trustee. In our view, trustworthiness represents the inherent, underlying potential for a trustee to be reliable, predictable, and fair. The distinction is based on the actual versus perceived intentions and motives of a trustee—the former being trustworthiness and the latter being trust. For our purposes, trust as a cognitive state is synonymous with the perception of trustworthiness.

The motivation for purchasing managers to be trusted by supplier representatives is driven in part by the interdependence between exchange partners. Given that the purchasing manager depends on the supplier representative for procurement needs that help fulfill some goals of the purchasing manager and her organization, the purchasing manager has an incentive to behave in a trustworthy manner. Purchasing managers may also have an interest in being trusted by supplier representatives in order to better deal with the large number of contingencies and noncontractibles that are part of any interfirm exchange relationship. When extraordinary responses are required on the part of the supplier, such as a rush delivery, supplier representatives that believe that the purchasing manager is not, or has not been, misrepresenting the situation would be inclined to make efforts beyond the letter of the contract.

Role and Role Autonomy

While a job (i.e., the set of task elements assigned to a single employee) is the component of an individual's work experience that is relatively fixed, formal, and derived from the structural properties of an organization, a role is the more emergent, dynamic, and socially defined component of the very same experience. Roles, in fact, contain "emergent task elements plus those elements of the job that are communicated to the job incumbent through the social system and maintained in that system" (Ilgen and Hollenbeck 1992, p. 174). A role can be identified as a "set of recurrent behaviors appropriate to a particular position in a social system" (Polzer 1995a, p. 495). A role then connects the person—his or her attributes, attitudes, and behaviors—to the organizational relations, systems, and structures that enable and constrain role behavior (Barley 1990). Roles are defined by the expectations[6] of role senders, i.e., the members of the role set with whom the role incumbent interacts (Katz and Kahn 1978).

The processes by which roles emerge have been described in terms of role taking and role making (Graen 1976). In role taking, the incumbent complies with the expectations established and communicated by members of the role set. From this perspective, a role delimits the decision-making latitude of the person performing a task (Griffin 1987, Griffin and McMahan 1994). Role making is a less passive process whereby incumbents actively influence the definition of their role in a way that is mutually satisfactory to members of the role set and to themselves. As a result of the two processes, roles are enacted and emerge as an outcome of both individual and organizational influences (Miner 1987, Welbourne et al. 1998).

Role making and role taking provide a compelling approach for highlighting the duality inherent in roles, much like two sides of a coin. On the one hand, roles shape behavior and act as norms defined by role senders—the constituents of the role set that express role expectations (in this study, supplier representatives and individuals in buyer-purchasing organizations). On the other hand, roles can be and are interpreted, sometimes in strategic ways, by role incumbents (Tsui et al. 1995). Taken together, role taking and role making suggest that incumbents experience a variable level of role autonomy in their organizations that goes from passive "custodianship" of the received role to "role innovation," where they are relatively free to actively change the goals and scope of their own role (Van Maanen and Schein 1979, Miner 1987). We define *role autonomy* as the degree of freedom role incumbents have in balancing the diverse expectations from their role set by devising appropriate actions and behaviors.

Because role autonomy is subject to both individual and organizational influences, we conceptualize it as a multidimensional concept. Organizational influences derive from a variety of sources, but we believe that structure and culture are among two of the most important. For this reason, we include functional influence as a way of capturing the effects of structure, and clan culture as a means of capturing the effects of culture, on role definition. In order to capture the individual influence on role, we include the individual's tenure in the organization. We elaborate on these three key components of role autonomy below.

We maintain that in an organizational context, boundary-spanners' role autonomy is defined by: (1) The level of influence that other internal functions have on boundary-spanners' decision-making discretion; (2) the level of experience, competence, and power accumulated over time working in the organization; and (3) the degree to which the organizational culture socializes boundary spanners to internalize collective goals and values. Direct influence from other organizational functions—what we refer to as *functional influence*—derives from interdependence between organizational units and implies the need to negotiate and coordinate decisions and actions. The freedom of purchasing managers is constrained by the amount of influence other units may exert upon them (Crozier and Friedberg 1977). A second way that purchasing managers' role autonomy is shaped is through on-the-job experience accrued over their *tenure* in the organization. In this case it is purchasing managers' knowledge of, and power within, the organization that gives them the freedom to make and uphold commitments.[7] Acculturation is the third means by which purchasing managers' role autonomy is enacted. While acculturation is less direct than functional influence, it acts at a more profound level. By internalizing the values and goals of an organization, a purchasing manager is able to act in a way that is consistent with the culture of the organization. Because *clan cultures* grant their members latitude in how they define their organizational roles, we view it as another relevant component of role autonomy.[8]

Prior research on clan control and behavioral discretion supports our conceptualization of clan cultures as an element of role autonomy (Floyd and Lane 2000, Zammuto and O'Connor 1992). Das and Teng argue "clan control is exercised when organizations do not specify task-related behaviors and outputs. Instead the focus is on developing shared values, beliefs, and goals among members so that appropriate behaviors will be reinforced and rewarded" (Das and Teng 2001, p. 259). Furthermore, "the underlying assumption of [clan] control is that people can ultimately determine their own behavior... Since there is no explicit restriction on members' behavior, more interpersonal respect and less mistrust are implied in [clan] control than in formal control" (Das and Teng 1998, p. 502).

This line of work makes a clear distinction between the organization's values, beliefs, and goals on the one hand and individual behaviors on the other, with clans constraining the former but not the latter. Not only does clan culture provide purchasing managers with a set of values and norms to guide their behavior, but also the leeway to act with autonomy in specific situations.

Taken together, we view functional influence, tenure, and clan culture as relevant components of purchasing managers' role autonomy. We argue that these elements influence the level of trust that supplier representatives place in purchasing managers.

Role Autonomy and Trust

We recognize that the inclination of purchasing managers to behave in a trustworthy manner is likely to vary. This raises the question of how purchasing managers granted role autonomy will use their discretion. We assert that more often than not, the granting of role autonomy will act as a self-fulfilling prophecy (Lewis and Weigert 1985). In other words, purchasing managers who are trusted with the stewardship of supplier relations will be more inclined to behave in a trustworthy manner than those who are closely monitored. This is consistent with broader evidence that when trust is extended to an individual, that person is more likely to behave in a trustworthy manner (Berg et al. 1995).

We predict that purchasing managers with greater role autonomy engender greater trust from supplier representatives than purchasing managers who are highly constrained in their boundary-spanning role. We argue that role autonomy influences trust because it provides the purchasing manager with an opportunity to display discretionary and competent behavior, thereby permitting the supplier representative to learn about the purchasing manager's trustworthiness. This is premised on the notion that the purchasing manager's true intentions and motives as an individual are made more visible when the veil of the purchasing manager's role is partially lifted. Purchasing managers who fulfill suppliers' positive expectations by simply adhering to role constraints reveal little about their own trustworthiness as individuals. Roles that are too constraining have the effect of masking the real intentions and capabilities of the person in a boundary-spanning role, making the placement of trust in the person difficult and potentially unjustified. This is consistent with the idea that "a strong organizational control system could inhibit the development of trust, because a trustee's actions may be interpreted as responses to that control rather than signs of trustworthiness" (Mayer et al. 1995, p. 727). When behavior is almost completely prescribed, the most relevant and salient target for trust is the organization, because the organization is the source of the boundary-spanner's regulated behavior. A more informative test of trust in a person acting in the role of purchasing manager is possible when the individual has the opportunity to exercise discretion in performing the role. Role autonomy allows supplier representatives to form attributions about purchasing managers as *individuals* as distinct from purchasing managers acting as agents performing prescribed organizational roles (Jones et al. 1961, Jones and Davis 1965).

Role autonomy is particularly important for dealing with the unforeseen contingencies, and associated adjustments, that arise in the context of industrial buyer-supplier relationships (Macneil 1980; Williamson 1975, 1985). Having the autonomy to resolve potential conflicts stemming from such unanticipated eventualities allows purchasing managers to exhibit discretionary behaviors that signal their trustworthiness. More specifically, by discovering mutually acceptable solutions for upholding the spirit of implicit commitments in the face of uncertainty, purchasing managers demonstrate fairness. Such *integrative* behavior on the part of the purchasing manager indicates fairness because it reflects a propensity to harmonize the interests of both parties. Another way that purchasing managers display discretionary behaviors is to take personal initiative and make independent decisions, instead of deferring to organizational rules and role constraints. Being *responsive* in this way allows the purchasing manager to keep promises made to her counterpart and be perceived as more predictable. Discretionary behavior is also reflected in a purchasing manager who uses her expertise and power to influence internal constituencies to accept and support the agreements reached with the supplier representative. This kind of *competent* behavior by the purchasing manager will be perceived as an indication of reliability by the supplier representative. In sum, the discretion to act in an integrative, responsive, and competent manner is what engenders interpersonal trust at the boundary. Below we explain how these discretionary behaviors relate functional influence, tenure, and clan culture to the level of trust that the supplier representative places in the purchasing manager.

Functional Influence

We define functional influence as the ability to get others to act as one intends by placing constraints on their behavior. This form of influence is based on functional interdependence, which creates the need for organizational subunits to coordinate activities (Thompson 1967).

The involvement of other functional areas in the decision making of boundary spanners represents the power to constrain actions and interactions with exchange partners. We argue that functional influence affects the ability of a purchasing manager to be responsive to the positive expectations of supplier representatives under conditions of uncertainty, thereby influencing the perceived predictability of purchasing managers (Strauss 1962, Anderson and Chambers 1985).

The moment of truth for purchasing managers to demonstrate their trustworthiness occurs when they are faced with an eventuality that does not alter the letter of the agreement but requires a modification in how the terms of the contract are to be fulfilled. Put differently, this is the issue of unforeseen contingencies or noncontractibles in the formal agreement (Williamson 1985). Such contingencies may occur when, for example, market demand changes, new product specifications are called for, or a new technology becomes available. Under these circumstances, purchasing managers require role autonomy to respond to these contingencies in order to balance the differing expectations coming from their own and from partner organizations.

The issue here is which party is going to bear what proportion of the cost of adaptation (Williamson 1985). Contradictory expectations between buyer and supplier organizations about how to adapt to change cause boundary spanners to experience role conflict, which we define as the "experience of contradictory, incompatible, or competing role expectations" (Rizzo et al. 1970). Constrained purchasing managers tend to act more as representatives of their own organizations instead of being gatekeepers of external interests as well (Friedman and Podolny 1992). Similarly, Conger and Kanungo (1988) argue that tight organizational constraints induce passive behavior, strict compliance to rules and requests coming from more powerful actors, and lack of initiative. When organizational constraints are high, it is more difficult for purchasing managers to make and consistently uphold obligations and commitments made to supplier representatives. Further, purchasing managers with less autonomy assume more rigid positions and will be inclined to bargain in more aggressive and distributive ways during negotiations (Raiffa 1982). These strategies have also been argued to lower the level of trust between individuals involved in a negotiation (Fisher and Ury 1981).

In contrast, purchasing managers with greater autonomy have the opportunity to pursue integrative agreements with supplier representatives, thereby enhancing their perception of fairness (Pruitt 1983, Walton et al. 1994). Being autonomous means that purchasing managers have the ability to be responsive not only to internal constituents, but also to the supplier representatives' expectations. From the supplier representative's perspective, being responsive means that the purchasing manager will consistently take into consideration the supplier's interest in adjusting to unanticipated changes. When functional influence is low, purchasing managers also have the power to play a buffering role by shielding the supplier from multiple demands from within the buyer organization. Thus, the buyer boundary spanner is able to act as a "variance absorber," smoothing out the stream of requests emanating from different functional areas (Asanuma 1989). From the perspective of the supplier representative, this will be viewed as stable and predictable behavior and will induce trust in the purchasing manager. Consequently, we predict that purchasing managers subject to greater influence from other functional units within their organization have less freedom to interpret their role and meet the positive expectations of supplier representatives. This leads us to our first hypothesis:

HYPOTHESIS 1. *There is a negative relationship between the degree of functional influence on the purchasing manager and the supplier representative's trust in the purchasing manager.*

Company Tenure

By company tenure we mean the length of time an individual has spent working in the organization. We maintain that the relationship between tenure and supplier trust in the purchasing manager has its basis in the informal power and knowledge that accrue to a long-tenured organizational member. Longer-tenured managers are, in most cases, more powerful because members of an organization typically gain influence and status as their tenure increases (Pfeffer 1982, 1992). One source of informal power is social capital, or the value derived from the network of individual relationships maintained in the organization (Coleman 1988). Over time, social capital accumulates as an individual forms more and more relationships and generates a series of obligations and expectations of reciprocity. These create "credit slips" that a manager can draw on in time of need (Coleman 1990). Relatedly, individuals with higher tenure are more likely to have learned about the social structure or network of the organization, whereas members new to the organization would possess a more limited understanding of the organizational social structure. This knowledge of the network, or "know-who" knowledge, is itself a form of power because it identifies the resources and coalitions that enable managers to carry out their agendas successfully (Krackhardt 1990).

Another source of power, expert power (French and Raven 1959), is premised on the increased working knowledge that individuals have acquired over time about their work, products, processes, and working style of the organization. As an individual's tenure in the organization increases, there are likely to be fewer others with a similar stock of knowledge, making it more valuable and the individuals possessing it more powerful (Crozier and Friedberg 1977). Purchasing managers with such knowledge have a better understanding of their role expectations and the manner in which to best maneuver in their role when faced with conflicting expectations (Kohli 1989). Along these lines, Jackson and Schuler note:

> Job tenure ... tends to be slightly negatively correlated with role ambiguity ... The longer one is in a job, the more information he or she obtains. An alternative explanation would be that employees who fail to obtain clarification over time leave the job (1985, p. 37).

Both the degree of informal power and knowledge of long-tenured purchasing managers are important precursors to the level of trust placed in them by supplier representatives. From the standpoint of a supplier representative, a competent purchasing manager is one who uses her power and knowledge to manage constituencies in the buyer organization to accept the agreements that she has reached with the supplier representative. Such purchasing managers will engender the trust of supplier representatives because they can be relied upon to follow through on their commitments and obligations. Put differently, long-tenured purchasing managers are able to use their influence and knowledge to ensure that they deliver on suppliers' positive expectations, whereas purchasing managers new to the organization are likely to wield less influence, making them less able to fulfill suppliers' positive expectations. Moreover, experienced purchasing managers possess the skills and abilities that give supplier representatives the assurance that their interests will be attended to competently. Accordingly, we predict that long-tenured purchasing managers, due to their knowledge and informal power, will have higher trust placed in them. Thus, we hypothesize:

HYPOTHESIS 2. *There is a positive relationship between the company tenure of a purchasing manager and the supplier representative's trust in the purchasing manager.*

Clan Culture

Organizational culture is defined as a set of values, beliefs, assumptions, and symbols that emerge "in a transactions framework that governs the expectations individuals may form in their dealings with others and that specifies the rights and obligations of each party" (Jones 1983, p. 458). Individuals embedded in strong organizational cultures will be influenced to a greater extent by internalized values and goals than by formal rules and procedures in determining appropriate actions for given situations. Such cultures can be found in organizations that have been described as *clans*, which are distinguished from other forms of organization by a reliance on socialization as the principal means of control (Ouchi 1980).

The clan form of organizing is most often observed under conditions of performance ambiguity—situations in which it is difficult to determine individual performance (Wilkins and Ouchi 1983). Under these circumstances, explicit monitoring and evaluation as a means to keep opportunism in check are increasingly difficult to implement due to the lack of precise, verifiable measures. Clans rely on the intense socialization of organizational members as an alternative mechanism for assuring individual performance when ambiguity is high. Socialization serves as an effective control mechanism because organizational members come to accept the organization's goals as their own. As a result, the inclination of members of a clan organization is to act in ways consistent with the goals and values of the organization, in the absence of close monitoring.

The ambiguity of performance measurement and goal congruence among organizational members, characteristic of clan organizations, holds important implications for the behavior of boundary spanners in such organizations. Since clan organizations do not precisely define tasks nor explicitly direct their members in the performance of activities, boundary spanners have a great deal of discretion over how to manage relationships with other organizations. Accordingly, we expect purchasing managers in clan organizations to be more responsive under conditions of uncertainty, and consequently to have a greater capacity to fulfill the positive expectations of supplier representatives.

The intense socialization of clan organizations also holds important implications for the behavior of boundary spanners in such organizations. Members of clan organizations are socialized to accept the idea that over time they will be dealt with equitably, thus allowing short-term imbalances in rewards that encourage cooperative behavior (Wilkins and Ouchi 1983). The internalization of these values orients boundary spanners toward long-term relationships, which facilitate the search for integrative solutions that involve payoffs that are extended across time. The internalization of equity as a value also stimulates the search for solutions that harmonize the interests of both parties in the relationship. Purchasing managers engaging in such integrative behaviors in their external relationships are likely to be perceived as fair and be trusted by the supplier representative. More formally,

HYPOTHESIS 3. *There is a positive relationship between the prevalence of a clan culture within the purchasing manager's organization and the supplier representative's trust in the purchasing manager.*

This prediction is consistent with an alternative explanation that views trust in the purchasing manager as an artifact of trust in the purchasing manager's *organization*, based on the notion that an organization can be viewed as trustworthy because of its internal organizing principles and practices. In particular, clan organizations, by virtue of their strong norms and intense socialization, promote goal congruence and reduce the inclination of organizational members to behave opportunistically. Along these lines, Ouchi observes that in clan organizations "a variety of social mechanisms reduce differences between individual and organizational goals and produce a strong sense of community" (1980, p. 136).

Clan organizations not only adhere to trust-based organizing practices and principles internally, but also apply those same ideals to relationships outside the organization (Dyer and Ouchi 1993). As Adler et al. argue, organizations find it difficult to "mix 'oil'—the high-trust, organic, innovation subunits within the firm—with 'water'—low trust relations with external suppliers outside the firm" (1999, p. 64). This suggests that organizations that adhere to clan principles by emphasizing goal congruence, creation of community, equitable dealing, and nonopportunistic behavior engender trust from their external exchange partners. Moreover, since it is the boundary spanners of a clan organization that carry out its trust-based routines with external exchange partners, the attribution of trust is likely to extend to those individuals as well. This reasoning is consistent with the notion of category-based processing (Fiske and Taylor 1991). Specifically, it has been argued that beliefs about the trustworthiness of a group influence the perception of trustworthiness of the individual members of that group (Williams 2001). Moreover, empirical research has demonstrated that trust in the organization and trust in the individual are positively related (Zaheer et al. 1998).

From this perspective, a supplier representative may trust a purchasing manager because the purchasing manager is a member of a trusted clan organization, rather than due to the purchasing manager's high role autonomy, suggesting an alternative causal mechanism to that of role autonomy in relating clan culture to trust in the purchasing manager.[9] Specifically, clan culture may influence trust in the purchasing manager indirectly through its effect on trust in the buyer organization. Accordingly, we propose as an alternative hypothesis that trust in the buyer organization mediates the relationship between clan culture and trust in the purchasing manager. Formally,

HYPOTHESIS 4. *The supplier representative's trust in the buyer organization positively mediates the relationship between the prevalence of a clan culture within the buyer organization and the supplier representative's trust in the purchasing manager.*

Controls

To assess the unique contribution of the theoretically derived constructs in explaining the level of trust placed in the boundary spanner, we control for interorganizational, organizational, interpersonal, and individual factors considered to be important determinants of trust.

Interorganizational. Trust in the partner organization is used as a control as well as a mediator in the test for Hypothesis 4. Trust in the partner organization is included as a control because it has been shown to be related to trust in the individual boundary spanner (Zaheer et al. 1998). This result is predicated on the reasoning that members of trusted organizations are assumed to behave in a way that is consistent with organization's practices and principles that engender trust. Accordingly, in the model we include a measure of the extent to which the supplier representative trusts the buyer organization (rather than the purchasing manager). In addition, it has been argued that a sense of *common fate* must develop between two interacting parties before they can share information freely and develop an open atmosphere conducive to trust (Powell 1996). Finally, we control for the *extendedness*, or the expected future duration of a relationship between two organizations. When the purchasing manager envisions an extended shadow of the future for the two organizations, we would predict that she is less likely to behave opportunistically and that the supplier representative will place greater trust in her for this reason. Extendedness deters opportunistic behavior and encourages trust because any potential gains from cheating are more than outweighed by the threat of future losses from retaliation (Heide and Miner 1992).

Organizational. Size of the buyer organization is included as a control because it is likely to be related to other organizational processes and outcomes that also influence trust in the purchasing manager. In this way, size serves as an important control for a number of omitted organizational variables.

Interpersonal. We control for the *past duration of the interpersonal relationship* with the purchasing manager because with a history of repeated interactions, the boundary spanners become increasingly confident of the other person's willingness and capacity to fulfill promises (Blau 1964). The *frequency of communication* has often been shown to be related to the development of trust, through repeated interaction that allows people to learn about another's trustworthiness (Lewicki and Bunker 1996, Rempel et al. 1985, Zand 1972). Boundary spanners that interact and exchange information more frequently get to know and understand each other better, which reduces the risk of misunderstandings and clarifies the expectations of both parties (Katz and Kahn 1978). Where expectations are clear, trust is likely to be higher. To the extent that frequency of communication is an indication of future potential interaction, it may influence the formation of trust in a manner similar to extendedness, by deterring opportunistic behavior.

Individual. Individuals trust others who appear more similar because it is assumed that similar others are likely to share common values and beliefs, a phenomenon that has been referred to as "characteristic-based trust" (Zucker 1986, McKnight et al. 1998). Accordingly, we use *age difference* to control for the effects of similarity in the ages of the boundary spanners on trust. Using the same reasoning, we also include a variable for the *same gender* as a control.

Methods

Research Design

The hypotheses developed in the preceding section predict a relationship between the level of trust supplier representatives place in purchasing managers and the degree of autonomy purchasing managers experience in interpreting their boundary-spanning roles. Our theoretical arguments rest on the notion that supplier trust in purchasing managers is based on the autonomy purchasing managers exercise to fulfill suppliers' positive expectations. Assessing the validity of such a theory requires examining the relationship between components of the purchasing manager's role autonomy on the one hand, and the level of trust supplier representatives place in purchasing managers, on the other. This suggests that the appropriate research design to test our predictions should include data from *both* the purchasing manager and the supplier representative who are primarily responsible for managing a given buyer-supplier relationship. In this study, purchasing managers reported on key components of role autonomy (functional influence, tenure, clan culture), and supplier representatives reported their level of trust in purchasing managers. While this research design poses the challenge of collecting and matching data from both members of a buyer-supplier relationship, there are significant methodological advantages associated with it. Most importantly, by using one knowledgeable source for measuring the dependent variable and another for measuring the independent variables, we avoid common method variance—a major difficulty confronting research on trust in interpersonal and interorganizational relationships.

Data Collection

We collected data for this study in four phases. We began by compiling a list of firms in the "Electronic and Other Electrical Equipment and Components" industry (SIC code 3600) by contacting the National Association of Purchasing Managers (NAPM). The association provided us with a list of members who are purchasing managers from firms operating in SIC code 3600. We then contacted 20 of these purchasing managers and conducted extensive semistructured interviews, lasting approximately 45 minutes each, in order to better understand the exchange context and the issues involved in the buying process.

In the second phase we identified 2,000 NAPM members who were eligible to participate in the study based on the following criteria: (1) Their firms purchase components, (2) they deal directly with supplier firms, and (3) they have a purchasing relationship with at least six suppliers of different components. We then mailed a letter to qualifying NAPM members, inviting them to participate in the study. In order to participate, purchasing managers were requested to identify and send to us the name and address of one specific supplier firm from which they source a key component.

Because there was the potential for purchasing managers to select supplier relations that were uniformly positive, as would likely be the case if the largest suppliers were selected, we used the following procedure to control for selection bias (Anderson and Narus 1990). Purchasing managers were instructed to select the *fourth*-largest firm (in terms of dollar amount of purchases made in the past year) and to identify the specific contact person (i.e., supplier representative) with whom they personally deal. A total of 268 purchasing managers agreed to participate by returning the requested information.

Testing for Nonresponse Bias. We suspected that nonresponse bias may have been a possibility, given the somewhat low participation rate of 13% (i.e., 268/2,000). To investigate the existence of systematic differences between our sample and the broader population of purchasing managers, we conducted a telephone survey of 100 randomly selected nonparticipants (i.e., purchasing managers who did not respond to our invitation to participate). Our analysis consisted of comparing the differences in the means of certain key variables, using t-tests, between the samples of participants and nonparticipants. We found no significant differences in the size of firms ($t = 1.10$), the length of the business relationship with the supplier companies ($t = -0.81$), or satisfaction with the buyer-supplier relationships ($t = -0.56$). To the extent that these variables are indicative of some key constructs in our study, we believe that our sample of purchasing managers is representative.

The third phase of data collection entailed mailing questionnaires to the 268 purchasing managers. At this time, we also sent a letter to the supplier contact persons identified by purchasing managers, inviting them to participate in the study. A different questionnaire sent along with the letter contained a label with the name and company of the purchasing manager who had identified them. The supplier representatives were instructed to complete their questionnaire, keeping in mind the particular buyer organization and purchasing manager identified. In the fourth and final phase of data collection we used Dilman's (1978) techniques for maximizing response rate, including: (1) Sending a reminder/thank-you postcard to all participants, (2) sending a second wave of questionnaires and reminder letters to nonrespondents, and (3) making follow-up telephone calls to any remaining nonrespondents.

A total of 187 purchasing managers and 184 supplier representatives completed and returned questionnaires, resulting in a final response rate of 69% (371/536). There were 149 complete dyads (i.e., instances where both the purchasing manager and supplier representative from the same buyer-supplier relationship returned responses) among the returned questionnaires. After eliminating questionnaires with missing values, there were 119 complete dyads (238 usable responses) available.

Questionnaire Instruments

We developed our questionnaire instruments using information obtained from the semistructured interviews and previous research. After having colleagues review the instruments and make suggestions for improving them, we pretested questionnaires with a group of local purchasing managers from the same industry. These responses were *not* included in the final data set used for analysis. Feedback from the pretest was used to make a final set of revisions to the questionnaire instrument.

Measures. The details of the measurement items and scales used to operationalize the theoretical constructs, along with their associated Cronbach alpha reliability values, are reported in Table 1. Reliability values of most constructs meet Nunnally's (1978) criterion of acceptability of 0.70 or above. The reliability values of the Clan culture (alpha = 0.69) and Common fate (alpha = 0.66) constructs are marginal. Means, standard deviations, and zero-order correlations among constructs are reported in Table 2. The development of our instrument for measuring trust in the purchasing manager warrants further discussion.

Trust in the Purchasing Manager. We developed a measure of the supplier representative's trust in the purchasing manager based on a measurement instrument of trust in close, interpersonal relations, created and validated by Rempel et al. (1985).[10] We adapted the original scale to make it more suitable for buyer-supplier relationships by modifying items measuring a faith component to also tap into fairness, consistent with our definition of trust. We also modified the referent of trust to refer to the supplier representative's "contact person" (i.e., the purchasing manager).

Construct Validity. In this study, we sought to assure the validity of our constructs by using multi-item measurement instruments that were previously validated. However, we recognize that previous evidence of a measurement instrument's construct validity does not necessarily guarantee that the instrument will prove to be valid in a different context. For this reason, we performed a confirmatory factor analysis to examine the validity of our measures. Using LISREL VIII (Jöreskog and Sörbom 1996), we estimated a measurement model corresponding to the theoretically derived factor structure of the measurement instruments. The measurement model contained seven latent variables (one for each multi-item measurement instrument) with their corresponding manifest (observed) variables assigned to them. This model was compared to a null model that assigned all manifest variables to a single latent variable. If the theoretically derived factor structure accurately represents the underlying data, the measurement model should produce a superior fit. The results of the confirmatory

Table 1 Measurement Items and Constructs Reliability

Items and Constructs[a]	Alpha	Source
Trust in the purchasing manager[b]	0.76	Adapted from Rempel et al. (1985)
1. My contact person has always been evenhanded in negotiations with me.		
2. My contact person may use opportunities that arise to profit at my expense. (R)		
3. I know how my contact person is going to act. S/he can always be counted on to act as I expect.		
4. Based on past experience, I cannot with complete confidence rely on my contact person to keep promises made to me. (R)		
5. My contact person is trustworthy.		
Functional influence	0.78	New scale
How much do individuals from the following functional areas in your business unit constrain your decisions about the business relationship with Supplier *X* during the *past year*?		
1. Manufacturing		
2. Engineering		
3. Research and development		
4. Quality		
5. Finance/accounting		
(Scale: Seven-point Likert scale ranging from (1) Not at all to (7) To a great extent. Measure: Items sum)[13]		
Organizational tenure	n.a.	
1. How long have you been working *for this company*? (Measure: Logarithm of years)		
Clan culture	0.69	Adapted from Moorman et al. (1993)
1. Our business unit is a very personal place. It is like an extended family. People seem to share a lot of themselves.		
2. The glue that holds our business unit together is loyalty and tradition. Commitment to this firm runs high.		
3. Our business unit emphasizes human resources. High cohesion and morale are most important.		
(Measures: 0 to 100 points based on how much each statement describes the respondent's company. Scale: Items sum)		
Trust in the partner organization[b]	0.77	Adapted from Rempel and Holmes (1986)
1. Customer *X* has always been evenhanded in its negotiations with us.		
2. Customer *X* may use opportunities that arise to profit at our expense. (R)		
3. Based on past experience, we cannot with complete confidence rely on Customer *X* to keep promises made to us. (R)		
4. We are hesitant to transact with Customer *X* when the specifications are vague. (R)		
5. Customer *X* is trustworthy.		
Common fate	0.66	Adapted from Anderson and Narus (1990)
1. A problem solved by Supplier *X* means a problem solved for our business unit.		
2. We view Supplier *X* as our ally against competition.		
3. We see our success as directly dependent upon the success of Supplier *X*.		
Extendedness of business relationship	0.93	Heide and Miner (1992)
1. Our business unit expects our relationship with Supplier *X* to last a long time.		
2. It is assumed that agreements with Supplier *X* will be renewed.		
3. Our business unit makes plans not only for the terms of individual purchases, but also for the long-term relationship with Supplier *X*.		

Note. [a]Unless indicated otherwise, items measured with a seven-point Likert scale ranging from (1) Strongly disagree to (7) Strongly agree; scale items averaged.

[b]From supplier representatives' questionnaires.

[c]From purchasing manager and supplier representatives' questionnaires.

(R) = reverse scored.

Table 1 (cont'd.)

Items and Constructs[a]	Alpha	Source
Size	n.a.	
1. Please let us know the number of full-time employees working in your company at this time. (Measure: Logarithm number of employees)		
Past duration of the relationship[c]	n.a.	
1. How long have you known the contact person at Customer/Supplier *X*? (Measure: Logarithm mean of purchasing manager and supplier representative responses)		
Frequency of communication[c]	0.74	New scale
On average, about how often did you use the following to communicate with your contact person during the *past year*?		
1. Telephone		
2. Fax		
3. Scheduled meetings		
4. Unscheduled meetings		
5. E-mail		
(Scale: 1 = never; 2 = yearly; 3 = quarterly; 4 = monthly; 5 = weekly; 6 = twice a week; 7 = daily. Measure: Sum of purchasing manager and supplier representative responses)		
Age difference[c]	n.a.	
1. How old were you on your last birthday? (Measure: Purchasing manager age − supplier representative age)		
Same gender[c]	n.a.	
1. Please indicate your gender. (Measure: 1 = same, 0 = different)		

Note. [a]Unless indicated otherwise, items measured with a seven-point Likert scale ranging from (1) Strongly disagree to (7) Strongly agree; scale items averaged.

[b]From supplier representatives' questionnaires.

[c]From purchasing manager and supplier representatives' questionnaires.

(R) = reverse scored.

Table 2 Constructs, Descriptive Statistics, and Zero-Order Correlations

Variables	Mean	s.d.	1	2	3	4	5	6	7	8	9	10	11
1. Trust in the purchasing manager	5.92	1.06											
2. Age difference	0.13	10.86	−0.00										
3. Same gender	0.71	0.46	−0.18	−0.11									
4. Size	6.70	2.12	0.00	−0.11	0.15								
5. Past duration of the relationship	1.37	0.75	0.15	0.06	0.17	−0.11							
6. Frequency of communication	33.35	7.46	0.06	−0.01	−0.07	0.04	−0.03						
7. Trust in the partner organization	5.54	1.04	0.64**	0.06	−0.15	−0.08	0.18	0.10					
8. Common fate	5.01	0.97	0.23*	0.03	0.01	−0.01	0.17	0.13	0.16				
9. Extendedness of business relationship	5.70	1.66	0.14	−0.05	−0.08	−0.13	0.03	0.04	0.07	0.21			
10. Functional influence	10.77	5.90	−0.22*	−0.25**	0.06	0.13	−0.30**	0.12	−0.13	−0.14	0.02		
11. Organizational tenure	1.85	0.89	0.27**	0.18	0.05	0.13	0.49**	−0.05	0.15	0.18	−0.16	−0.23*	
12. Clan culture	59.08	47.01	0.21*	0.15	0.06	−0.32**	0.09	0.04	0.25*	0.01	0.09	0.01	0.02

Note. **Correlation is significant at the 0.01 level (two-tailed); *Correlation is significant at the 0.05 level (two-tailed); $N = 119$.

(R) = reverse scored.

factor analysis indicate that the null model yielded a χ^2 (189 df) = 866.21, $p < 0.000$, while the measurement model yielded a χ^2 (174 df) = 193.03, $p < 0.15$. The difference in chi-square (15 df) of 673.18 is statistically significant at $p < 0.01$. Moreover, standard fit indices for the measurement model indicate a satisfactory fit of the model with the data. Furthermore, the completely standardized lambdas for individual measurement items show high loadings onto their corresponding theoretical factors and are all statistically significant at $p < 0.01$. We view these results as offering support for the unidimensionality of the constructs and confirming the theoretically derived factor pattern of the measurement model.[11]

Results

We use hierarchical regression analysis to test Hypotheses 1–3. The dependent variable in the models is *trust in the purchasing manager* as assessed by the supplier representative. The first model includes the control variables. We then add each role autonomy variable (*clan culture, functional influence*, and *tenure*) one at a time to test their unique influence on the supplier representatives' *trust in purchasing managers*. The three role autonomy variables, along with the control variables *common fate, extendedness of the business relationship*, and *size* are all measured using data from the purchasing managers. The remaining controls (*gender similarity, age difference, past duration of the interpersonal relationship, frequency of communication*) are measured using data from both the purchasing manager and supplier representative (Table 1 provides the details for how the data from the two surveys were combined to compute specific variables).

The results of the hierarchical regression analysis are reported in Table 3. We focus on the estimates in the final model (Model V). We find evidence in support of the first hypothesis, that *functional influence* is negatively related to *trust in the purchasing manager*. The coefficient is negative and statistically significant ($\beta = -0.028$, Std. Error = 0.014; $p < 0.05$). Supplier representatives' trust in purchasing managers is greater when purchasing managers are subject to relatively little influence from other functional areas. The results also support the second hypothesis that purchasing manager *tenure* is positively associated with *trust in the purchasing manager* ($\beta = 0.261$, Std. Error = 0.103; $p < 0.05$). This finding is consistent with our argument that more senior purchasing managers have accumulated the knowledge and power necessary to carry through on their commitments and obligations to supplier representatives. Hypothesis 3, which predicts that *clan culture* is positively related to *trust in the purchasing manager*, was not borne out by the data. Although the coefficient is initially positive and statistically significant in Model IV ($\beta = 0.006$, Std. Error = 0.002; $p < 0.05$), the coefficient turns nonsignificant with the introduction of the *trust in the partner organization* variable used to test the mediation Hypothesis 4.

Hypothesis 4, which predicted that the relationship between *clan culture* and *trust in the purchasing manager* is mediated by *trust in the buyer organization*, is tested using the three-step procedure described by Barron and Kenney (1986). We first estimate the direct effect between *clan culture* and *trust in the purchasing manager*. Next we include the mediator variable *trust in the partner organization*. Finally, we estimate the relationship between *clan culture* and *trust in the partner organization*. Evidence in support of a mediation relationship requires that: (1) The relationship between *clan culture* and *trust in the purchasing manager* is initially statistically significant, (2) this relationship turns nonsignificant with the introduction of the mediator variable *trust in the partner organization*, (3) the relationship between the mediator variable (*trust in the partner organization*) and *trust in the purchasing manager* is statistically significant, and (4) the relationship between *clan culture* and the mediator variable (*trust in the partner organization*) is statistically significant. All four of these conditions are met. As shown in Table 3, Model V, when the mediator variable, *trust in the partner organization*, is introduced the *clan culture* variable becomes nonsignificant while the mediator variable is statistically significant.[12] To test the relationship between *clan culture* and *trust in the partner organization* we estimated a separate regression equation which included as controls the variables *size, past duration of the relationship, communication frequency, common fate*, and *extendedness of the relationship*. The relationships between *clan culture* and *trust in the partner organization* is positive and statistically significant ($\beta = 0.005$, Std. Error = 0.002; $p < 0.01$). This pattern of findings supports a full mediation relationship between *clan culture* and *trust in the purchasing manager*.

Discussion

Perhaps the most important contribution of this research is to embed trust more deeply in the organizational context. We proposed that in the context of buyer-supplier relations, where interactions between individuals represent "institutionalized personal relations" (Seligman 1997, p. 18), the nature and content of the relationship

Table 3 Results of Multiple Regression Analysis

	Model				
	I	II	III	IV	V
Independent Variables	β (Std. Error)	β (Std. Error)	β (Std. Error)	β (Std. Error)	β (Std. Error)
Controls					
Same gender	−0.491*	−0.596*	−0.535*	−0.496*	−0.240
	(0.223)	(0.218)	(0.216)	(0.212)	(0.176)
Age difference	−0.028	−0.006	−0.010	−0.014	−0.012
	(0.009)	(0.009)	(0.009)	(0.009)	(0.007)
Size	0.035	0.070	0.084	0.059	0.043
	(0.048)	(0.049)	(0.048)	(0.048)	(0.039)
Past duration of the relationship	0.230†	0.222†	0.143	−0.024	−0.130
	(0.136)	(0.132)	(0.135)	(0.151)	(0.124)
Frequency of communication	0.003	0.001	0.003	0.006	0.007
	(0.013)	(0.013)	(0.013)	(0.013)	(0.010)
Common fate	0.202*	0.207*	0.184†	0.153	0.080
	(0.104)	(0.101)	(0.100)	(0.099)	(0.081)
Extendedness of business relationship	0.062	0.051	0.056	0.082	0.073
	(0.060)	(0.058)	(0.058)	(0.057)	(0.047)
Role autonomy					
Clan culture		0.005**	0.006**	0.006**	0.003
		(0.002)	(0.002)	(0.002)	(0.002)
Functional influence			−0.036*	−0.034*	−0.028*
			(0.017)	(0.017)	(0.014)
Organizational tenure				0.293*	0.261*
				(0.126)	(0.103)
Mediator					
Trust in the partner organization					0.560***
					(0.077)
R^2	0.120	0.179	0.213	0.253	0.511
ΔR^2	0.120†	0.059**	0.033*	0.040**	0.258***
Adjusted R^2	0.061	0.115	0.142	0.253	0.457
ΔF	2.02*	7.29***	4.30*	5.36*	52.39***

Note. †$p < 0.10$; *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.
Dependent variable: *Trust in the purchasing manager*, $N = 119$.
Unstandardized beta coefficients.

between purchasing managers and their own organizations affects the level of trust that supplier representatives place in purchasing managers. The theoretical link between internal organizational constraints and external trust is captured by the idea of role autonomy. At the boundary of organizations, roles help to determine what purchasing managers should do to meet the positive expectations of supplier representatives and their own organizations. We argued that when purchasing managers interpret and enact their role in a way that provides autonomy from organizational constraints, they are better able to meet the positive expectations of their external counterparts by being more integrative, responsive, and competent. As a result, these more-autonomous purchasing managers elicit higher levels of trust.

The findings provide broad support for these arguments. The results suggest that supplier representatives' trust in purchasing managers is higher when: (1) Purchasing managers are free from constraints imposed by other functional areas; (2) the tenure of purchasing managers is longer; and indirectly (3) the culture of purchasing managers' organizations is more supportive of clan values and norms. In short, these findings suggest that boundary spanners with greater autonomy to manage interorganizational relations are better able to cultivate trust from their counterparts. Freed from the constraint of strictly adhering to rigid rules and requirements of their own organizations, boundary-spanning agents have greater latitude in upholding commitments to partner organizations and fulfilling the positive expectations of their representatives.

Although the results are largely consistent with the hypotheses, the results for the influence of clan culture on trust in the purchasing manager warrant additional comment. We had proposed alternative paths through which the culture of the buyer organization might relate to the level of trust that supplier representatives place in purchasing managers. Specifically, the hypothesized direct effect between clan culture and trust did not materialize, while the indirect effect through trust in the buyer organization was supported in the data. The implication of this finding is that the effect of clan culture on trust in the purchasing manager is due to the organization's reputation for fair dealing, rather than the role autonomy of the purchasing manager. The finding is consistent with the idea that trust-based practices and routines of clan organizations engender trust from external parties, and these parties extend their perception of the clan organization's trustworthiness to its members.

Based on the overall findings, we suggest that for trust to develop in individuals, the roles that form social structures must leave some room for interpretation. This latitude is crucial for the creation of trust. As Seligman points out, trust is "some sort of belief in the goodwill of the other, given the opaqueness of other's intentions and calculations. The opaqueness... rests precisely on that aspect of alter's [the purchasing manager's] behavior that is beyond the calculable attributes of role fulfillment; if it were otherwise, alter's [the purchasing manager's] actions would not be unknown but assessable within the framework of the defining system of role expectations and hence reflect confidence and not trust" (1997, p. 43). When roles are not completely defined, behavior is more subject to free will, and trust in individuals becomes more meaningful and relevant. Trust emerges when the space opened by role autonomy is filled by discretionary behaviors. In other words, agents involved in interorganizational relationships need to be relatively free to be trusted.

Limitations and Suggestions for Future Research

While we believe that our findings have advanced our understanding of trust in the organizational context, it is important to highlight features of our research design that may limit the generalizability of our findings. While our results are consistent with the theoretical arguments presented, we have theorized about mediating mechanisms that were not specifically measured and tested. Specifically, discretionary behaviors—integrativeness, responsiveness, and competence—that we suggested as mediating between role autonomy and trust could be the subject of future research efforts.

Moreover, the cross-sectional nature of our study precludes us from investigating possible endogenous effects in our model. It would be useful to conduct a longitudinal study that evaluates the extent to which trust in individuals co-evolves along with, and influences, features of the organizational context. For instance, a manager who has earned a reputation for being trustworthy may be granted greater role autonomy. Examining some of these reciprocal relationships would provide a more comprehensive understanding of the link between trust in individuals and the organizational context.

Aside from our specific empirical contribution, our research also advances the trust literature by shifting the focus away from interpersonal, and toward contextual, effects on trust. The focus on role autonomy is important because it provides a conceptual vehicle for making the micro-macro link between trust in individuals and the organizational context. Apart from the contribution to the trust literature, our research also advances thinking in the area of organizational roles by crystallizing the role-making and role-taking ideas into the concept of role autonomy. Moreover, while the literature on organizational roles primarily focuses on the consequences for the role incumbent, our research develops a broader set of implications of role autonomy, in particular for those interacting with the role incumbent.

Further, the concept of role autonomy creates opportunities to explore new areas of trust research such as relationships at different hierarchical levels or across different types of organizational boundaries. In addition, our conceptual approach can be used to deepen and complement existing research. For example, a growing body of research examines trust in organizational authorities and leaders. Some of this research has shown that it is important for leaders to be trusted when performance ambiguity exists or behaviors cannot be observed or controlled (Dirks 2000). Our findings are entirely consistent with these results, and our concept of role autonomy provides a generalizable theoretical rationale that integrates these findings with other similar research (Brockner et al. 1997, Robinson 1996).

While our initial exploration of purchasing managers' role autonomy and the level of trust placed in them by supplier representatives has yielded several important insights, it also raises additional questions for future research to address. Chief among these are the boundary conditions of our theory. We suspect that excessively high levels of role autonomy may be dysfunctional for the development of trust, particularly where organizational values are weakly developed and poorly internalized by organizational members. In such circumstances, boundary spanners may be tempted to behave in ways that further their own self-interest rather than that of the organization and that of its exchange partner. Alternatively, when the organization's values embody, tolerate, or even promote opportunism, role autonomy may be less relevant. In this case, agents that are role-bound would presumably behave opportunistically, as would agents with autonomy that have internalized the organization's values.

Summary

While role constraints may produce consistent behavior on the part of organizational agents, paradoxically such constraints preclude an assessment of the trustworthiness of the individual role occupant. We have argued and empirically demonstrated that for trust to emerge, individual discretion is critical. Role autonomy is what allows individuals in organizational contexts to exercise discretion in a way that conveys their underlying motives and intentions, highlighting the fact that trust relations are embedded in impersonal organizational structures and routines.

Acknowledgments

The authors are grateful to Stuart Albert, Kurt Dirks, Mark Fichman, Joe Galaskiewicz, Paul Ingram, Tanya Kostova, David Krackhardt, Ray Reagans, Peter Roberts, Henry Tosi, Sri Zaheer, three anonymous reviewers, and Senior Editor Anne Miner for their insightful comments and suggestions. Clearly, they are not responsible for the contents of this manuscript. This research was supported by grants to the first author from the Organization and HRM Department SDA, Bocconi University School of Management, and to the second author from the International Program Development (IPD) Office of the Carlson School of Management, University of Minnesota. All financial support is gratefully acknowledged. The National Association of Purchasing Managers kindly provided access to membership data.

Endnotes

[1]For instance, supplier expectations may include a stable and long-term relationship along with a price that ensures a fair profit margin. In contrast, the accounting function in the buyer organization may expect the lowest possible price and favorable payment terms, the quality control group might expect zero defects, and manufacturing may want components delivered just-in-time.

[2]We adopt the perspective of the supplier representative because, in the context of this study, the pressure on supplier boundary spanners to minimize the risk to their organization is greater than that on the purchasing managers. This occurs because the degree to which supplier boundary spanners have committed assets customized to the buyer's requirements is far higher than that of the buyer. In this study we find that suppliers report significantly higher levels of asset specificity than buyers ($t = 3.12$; $p < 0.005$). Higher asset specificity suggests a higher degree of vulnerability and, consequently, a higher potential for opportunistic exploitation (Williamson 1975). Under these circumstances, trust may serve as an effective safeguard to lower the risk of opportunism (Ghoshal and Moran 1996). Substantively, the focus on supplier representatives' trust in purchasing managers is also important. With more open information exchange, supplier boundary spanners are often confronted with the risk that proprietary information will be leaked to direct competitors or exploited by buyer counterparts in negotiating lower prices. While in this case suppliers are more vulnerable than buyers, it is possible for the reverse to hold in other contexts.

[3]Hereafter, we use the term trust to specifically refer to trust in the purchasing manager rather than trust in the buyer organization.

[4]Predictability captures the degree to which an agent acts consistently. This is distinguished from reliability, which reflects an agent's competence (Barber 1983).

[5]In addition to its use by Bromiley and Cummings (1995), fairness as a trust dimension has a well-established precedent in the literature. Specifically, Gabarro (1978) include a dimension of fairness in their trust scale. Building on this work, Robinson (1996) also incorporates fairness as a dimension of trust. In addition, Cummings and Bromiley's (1995) scale validation provides empirical evidence in support of fairness as a dimension of trust.

[6]Role expectations are "beliefs and attitudes held by members of the role set regarding what behaviors are appropriate for the person in the role" (Polzer 1995b, p. 499).

[7]We view commitments as the explicit and implicit promises made to members of the role set. As a result of commitments, expectations arise that the role occupant will uphold promises made.

[8]We recognize that not all types of culture result in increased role autonomy for its members. For example, bureaucratic organizations that value adherence to rules, hierarchy, and the chain of command may actually constrain rather than enhance the role autonomy of its members.

[9]We gratefully acknowledge an anonymous reviewer for suggesting this idea.

[10]Recently, several carefully designed and validated instruments for measuring trust specifically in interorganizational relations have been developed (see, for example, Currall and Judge 1995 or Cummings and Bromiley 1996). Unfortunately, our data collection efforts commenced before these instruments were available. For this reason we elected to use the Rempel et al. (1985) trust measurement instrument.

[11]Complete details of the confirmatory factor analysis are available from the authors upon request.

[12]The results of the mediation analysis also indicate a strong correlation between trust in the purchasing manager and trust in the partner organization ($r = 0.64$). This raises the question of whether supplier representatives can meaningfully distinguish between the two forms of trust. Our results support the assertion that the two forms of trust are, in fact, distinct. If indeed supplier representatives are confounding the constructs, then we should observe a similar pattern of relationships with other variables. Put differently, organizational tenure and functional influence should be correlated to both forms of trust. As the correlation matrix indicates (Table 2), the relationship between trust in the partner organization and both organizational tenure and functional influence are nonsignificant. Moreover, previous research

has also established the discriminant validity of trust in the purchasing manager and trust in the partner organization (Zaheer et al. 1998).

[13]We compute the functional influence scale using a sum in order to capture the aggregate level of influence faced by a purchasing manager from throughout the organization. Computing the scale based on the mean of the items would not accurately indicate the total level of influence faced, but rather the average level of influence. Functional influence should be greater as the number of functional areas exerting pressure on the purchasing manager increases, even when the influence of a given area is relatively low. A mean-based scale would reflect lower average influence, rather than higher aggregate influence, of a relatively low influence area.

References

Adams, S. J. 1976. The structure and dynamics of behavior in organizational boundary roles. M. D. Dunnette, ed. *Handbook of Industrial and Organizational Psychology.* Rand McNally, Chicago, IL, 1175–1199.

Adler, P. S., B. Goldoftas, D. I. Levine. 1999. Flexibility versus efficiency? A case study of model changeovers in the Toyota production system. *Organ. Sci.* **10**(1) 43–68.

Aldrich, H. E., D. Herker. 1977. Boundary spanning roles and organization structure. *Acad. Management Rev.* **2**(2) 217–230.

Anderson, E., B. Weitz. 1989. Determinants of continuity in conventional industrial channel dyads. *Marketing Sci.* **8**(4) 310–323.

Anderson, J. C., J. A. Narus. 1990. A model of distributor firm and manufacturing firm working partnerships. *J. Marketing* **54**(January) 42–58.

Anderson, P. F., T. M. Chambers. 1985. A reward/measurement model of organizational buying behavior. *J. Marketing* **49**(Spring) 7–23.

Asanuma, B. 1989. Manufacturer-supplier relationships in Japan and the concept of relation specific skill. *J. Japanese and Internat. Econom.* **30** 1–30.

Barber, B. 1983. *The Logic and Limits of Trust.* Rutgers University Press, New Brunswick, NJ.

Barley, S. R. 1990. The alignment of technology and structure through roles and networks. *Admin. Sci. Quart.* **35** 61–103.

Baron, R. M., D. A. Kenny. 1986. The moderator-mediator variable distinction in social psychological research: Conceptual, strategic, and statistical considerations. *J. Personality and Soc. Psych.* **51**(6) 1173–1182.

Berg, J., J. Dickhaut, K. McCabe. 1995. Trust, reciprocity, and social history. *Games and Econom. Behavior* **10** 122–142.

Biddle, B. J. 1979. *Role Theory: Expectations, Identities and Behaviors.* Academic Press, New York.

Blau, P. M. 1964. *Exchange and Power in Social Life.* John Wiley, New York.

Brockner, J., P. A. Siegel, J. P. Daly, T. Tyler, C. Martin. 1997. When trust matters: The moderating effect of outcome favorability. *Admin. Sci. Quart.* **42** 558–583.

Bromiley, P., L. L. Cummings. 1995. Transaction costs in organizations with trust. R. Bies, B. Sheppard, R. Lewicki, eds. *Research on Negotiations in Organizations*, Vol. 5. JAI Press, Greenwich, CT, 219–247.

Butler, J. K. 1991. Toward understanding and measuring conditions of trust: Evolution of a conditions of trust inventory. *J. Management* **17**(3) 643–663.

Coleman, J. S. 1988. Social capital in the creation of human capital. *Amer. J. Sociology* **94**(Supplement) S95–S120.

——. 1990. *Foundations of Social Theory.* The Belknap Press, Cambridge, MA.

Conger, J. A., R. N. Kanungo. 1988. The empowerment process: Integrating theory into practice. *Acad. Management Rev.* **13** 471–482.

Crozier, M., E. Friedberg. 1977. *L'acteur et le système. Les contraintes de l'action collective.* Editions du Seuil, Paris, France.

Cummings, L. L., P. Bromiley. 1996. The Organizational Trust Inventory (OTI). Development and validation. R. M. Kramer, T. R. Tyler, eds. *Trust in Organizations.* Sage, Thousand Oaks, CA, 302–330.

Currall, S. C., T. A. Judge. 1995. Measuring trust between organizational boundary role persons. *Organ. Behavior and Human Decision Processes* **64** 151–170.

Das, T. K., B. Teng. 1998. Between trust and control: Developing confidence in partner cooperation in alliance. *Acad. Management Rev.* **23**(3) 491–515.

——, ——. 2001. Trust, control, and risk in strategic alliances: An integrated framework. *Organ. Stud.* **22**(2) 251–283.

Dilman, D. A. 1978. *Mail and Telephone Surveys: The Total Design Method.* Wiley, New York.

Dirks, K. T. 2000. Trust in leadership and team performance: Evidence from NCAA basketball. *J. Appl. Psych.* **85** 1004–1012.

Dyer, J. H., W. G. Ouchi. 1993. Japanese-style partnerships: Giving companies a competitive edge. *Sloan Management Rev.* **35** (Fall) 51–63.

Fisher, R., W. Ury. 1981. *Getting to Yes: Negotiating an Agreement Without Giving in.* Houghton Mifflin, Boston, MA.

Fiske, S. T., S. E. Taylor. 1991. *Social Cognition*, 2nd ed. McGraw-Hill, Inc., New York.

Floyd, S. W., P. J. Lane. 2000. Strategizing throughout the organization: Managing role conflict in strategic renewal. *Acad. Management Rev.* **25**(1) 154–177.

French, J. R. P., Jr., B. Raven. 1959. The bases of social power. D. Cartwright, ed. *Studies in Social Power.* University of Michigan, Institute of Social Research, Ann Arbor, MI, 150–167.

Friedman, R. A., J. Podolny. 1992. Differentiation of boundary spanning roles: Labor negotiations and implications for role conflict. *Admin. Sci. Quart.* **37**(March) 28–47.

Fukuyama, F. 1995. *Trust: The Social Virtues and the Creation of Prosperity.* Free Press, New York.

Gabarro, J. J. 1978. The development of trust, influence, and expectations. A. G. Athos, J. J. Gabarro, eds. *Interpersonal Behavior: Communication and Understanding in Relationships.* Prentice-Hall, Englewood Cliffs, NJ.

Ghoshal, S., P. Moran. 1996. Bad for practice: A critique of transaction cost theory. *Acad. Management Rev.* **21**(1) 13–47.

Graen, G. 1976. Role making processes within complex organizations. M. D. Dunnette, ed. *Handbook of Industrial and Organizational Psychology.* Rand McNally, Chicago, IL, 1201–1245.

Griffin, R. W. 1987. Toward an integrated theory of task design. B. Staw, L. L. Cummings, eds. *Research in Organizational Behavior*, Vol. 9. JAI Press, Greenwich, CT, 79–120.

——, G. C. McMahan. 1994. Motivation through job design. J. Greenberg, ed. *Organizational Behavior. The State of the Science*. Lawrence Erlbaum Associates, Hillsdale, NJ, 23–43.

Heide, J. B., A. S. Miner. 1992. The shadow of the future: Effects of anticipated interaction and frequency of contact on buyer-seller cooperation. *Acad. Management J.* **35**(2) 265–291.

Ilgen, D. R., J. R. Hollenbeck. 1992. The structure of work: Job design and roles. M. Dunette, L. Hough, eds. *Handbook of Industrial and Organizational Psychology*, Vol. 2. Consulting Psychological Press, Palo Alto, CA, 165–207.

Jackson, S. E., R. S. Schuler. 1985. A meta-analysis and conceptual critique of research on role ambiguity and role conflict in work settings. *Organ. Behavior and Human Decision Processes* **36** 16–55.

Jones, E. E., K. E. Davis. 1965. From acts to dispositions: The attribution process in person perception. L. Berkowitz, ed. *Advances in Experimental Social Psychology*, Vol. 2. Academic Press, New York.

——, ——, K. J. Gergen. 1961. Role playing variations and their informational value for person perception. *J. Abnormal and Soc. Psych.* **63** 302–310.

Jones, G. R. 1983. Transaction costs, property rights, and organizational culture: An exchange perspective. *Admin. Sci. Quart.* **28** 454–467.

Jöreskog, K. G., D. Sörbom. 1996. *LISREL 8: User's Reference Guide*. Scientific Software International, Chicago, IL.

Kahn, R. H., D. M. Wolfe, R. Quinn, D. J. Snoek. 1964. *Organizational Stress: Studies in Role Conflict and Ambiguity*. Wiley, New York.

Katz, D., R. L. Kahn. 1978. *The Social Psychology of Organizations*. Wiley & Sons, New York.

Kohli, A. 1989. Determinants of influence in organizational buying: A contingency approach. *J. Marketing* **53**(July) 50–65.

Krackhardt, D. 1990. Assessing the political landscape: Structure, cognition and power in organizations. *Admin. Sci. Quart.* **35**(June) 342–369.

Lewicki, R. J., B. B. Bunker. 1996. Developing and maintaining trust in work relationship. R. M. Kramer, T. R. Tyler, eds. *Trust in Organizations*. Sage, Thousand Oaks, CA, 114–139.

Lewis, J. D., A. J. Weigert. 1985. Trust as a social reality. *Soc. Forces* **63**(4) 967–985.

Macneil, I. R. 1980. *The New Social Contract*. Yale University Press, New Haven, CT.

Mayer, R. C., J. H. Davis, F. D. Schoorman. 1995. An integrative model of organizational trust. *Acad. Management Rev.* **20**(3) 709–734.

McKnight, D. H., L. L., Cummings, N. L. Chervany. 1998. Initial trust formation in new organizational relationships. *Acad. Management Rev.* **23**(3) 473–490.

Miner, A. S. 1987. Idiosyncratic jobs in formalized organizations. *Admin. Sci. Quart.* **32** 327–351.

Moorman, C., R. Deshpandé, G. Zaltman. 1993. Factors affecting trust in market research relationships. *J. Marketing* **57**(January) 81–101.

Nunnally, J. 1978. *Psychometric Theory*. McGraw Hill, New York.

Ouchi, W. G. 1980. Markets, bureaucracies, and clans. *Admin. Sci. Quart.* **25** 121–141.

Pfeffer, J. 1982. *Organizations and Organization Theory*. Pitman, Boston, MA.

——. 1992. *Managing with Power. Politics and Influence in Organizations*. Harvard Business School Press, Boston, MA.

Polzer, J. T. 1995a. Role. N. Nicholson, ed. *Encyclopedic Dictionary of Organizational Behavior*. Blackwell Publishers Inc., Cambridge, MA, 495–498.

——. 1995b. Role set. N. Nicholson, ed. *Encyclopedic Dictionary of Organizational Behavior*. Blackwell Publishers Inc., Cambridge, MA, 499–500.

Powell, W. W. 1996. Trust-based forms of governance. R. M. Kramer, T. R. Tyler, eds. *Trust in Organizations. Frontiers of Theory and Research*. Sage, Thousand Oaks, CA, 51–67.

Pruitt, D. G. 1983. Achieving integrative agreements. M. H. Bazerman, R. J. Lewicki, ed. *Negotiating in Organizations*. Sage Publications, Beverly Hills, CA, 35–50.

Raiffa, H. 1982. *The Art and Science of Negotiation*. Harvard University Press, Cambridge, MA.

Rempel, J. K., J. G. Holmes, M. P. Zanna. 1985. Trust in close relationships. *J. Personality and Soc. Psych.* **49** 95–112.

Ring, P. S., A. H. Van de Ven. 1994. Developmental processes of cooperative interorganizational relationships. *Acad. Management Rev.* **19**(1) 90–118.

Rizzo, J. R., R. J. House, S. I. Lirtzman. 1970. Role conflict and ambiguity in complex organizations. *Admin. Sci. Quart.* **15** 150–163.

Robinson, S. L. 1996. Trust and breach of the psychological contract. *Admin. Sci. Quart.* **41** 574–599.

Rousseau, D. M., S. B. Sitkin, R. S. Burt, C. Camerer. 1998. Not so different after all: A cross-discipline view of trust. *Acad. Management Rev.* **23**(3) 393–404.

Seligman, A. B. 1997. *The Problem of Trust*. Princeton University Press, Princeton, NJ.

Shapiro, S. P. 1987. The social control of impersonal trust. *Amer. J. Sociology* **93**(3) 623–658.

Spekman, R. E. 1979. Influence and information: An exploratory investigation of the boundary role person's basis of power. *Acad. Management J.* **22** 104–117.

Strauss, G. 1962. Tactics of lateral relationship: The purchasing agent. *Admin. Sci. Quart.* **7**(2) 161–186

Thompson, J. D. 1967. *Organizations in Action*. McGraw-Hill, New York.

Tsui, A. S., S. J. Ashford, L. St. Clair, K. R. Xin. 1995. Dealing with discrepant expectations: Response strategies and managerial effectiveness. *Acad. Management J.* **38** 1515–1543.

Van den Bos, K., H. A. M. Wilke, E. A. Lind. 1998. When do we need procedural fairness? The role of trust in authority. *J. Personality and Soc. Psych.* **75** 1449–1458.

Van Maanen, J., E. H. Schein. 1979. Toward a theory of organizational socialization. B. M. Staw, ed. *Research in Organizational Behavior*, Vol. 1. JAI Press, Greenwich, CT, 209–264.

Walton, R. E., J. E. Cutcher-Gershenfeld, R. B. McKersie. 1994. *Strategic Negotiations*. Harvard Business School Press, Boston, MA.

Welbourne, T. M., D. E. Johnson, A. Erez. 1998. The role-based performance scale: Validity analysis of a theory-based measure. *Acad. Management J.* **41** 540–555.

Wilkins, A. L., W. G. Ouchi. 1983. Efficient cultures: Exploring the relationship between culture and organizational performance. *Admin. Sci. Quart.* **28** 468–481.

Williams, M. 2001. In whom we trust: Group membership as an affective context for trust development. *Acad. Management Rev.* **26** 377–396.

Williamson, O. E. 1975. *Markets and Hierarchies*. The Free Press, New York.

——. 1985. *The Economic Institutions of Capitalism. Firms, Markets, Relational Contracting*. The Free Press, New York.

Zaheer, A., B. McEvily, V. Perrone. 1998. Does trust matter? Exploring the effects of interorganizational and interpersonal trust on performance. *Organ. Sci.* **9**(2) 141–159.

Zammuto, R. F., E. J. O'Connor. 1992. Gaining advanced manufacturing technologies benefits: The roles of organizational design and culture. *Acad. Management Rev.* **17**(4) 701–728.

Zand, D. E. 1972. Trust and managerial problem solving. *Admin. Sci. Quart.* **17** 229–239.

Zucker, L. G. 1986. Production of trust: Institutional sources of economic structure, 1840–1920. B. M. Staw, L. L. Cummings, eds. *Research in Organizational Behavior*, Vol. 8. JAI Press, Inc., Greenwich, CT, 53–111.